STA 3402 – Machine Learning & Neural Networks

Lecture 5 – Factor Analysis of Mixed Data (FAMD), Customer Segmentation with FAMD + HCPC, and Tutorials 1 & 2 (MCA, FAMD)

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Full study notes from the lecture recording, the FAMD lecture note, the Customer Segmentation lab, Tutorial 1 (MCA), Tutorial 2 (FAMD) and the R code (simple English).

All outputs and plots come from actually running the R code (FactoMineR 2.9, factoextra 1.0.7).

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1. Class notes

- **Homework (USArrests clustering):** the deadline was 5.00 pm. There were only **36 submissions** out of about 41 students. The submission link was hidden after the deadline – the lecturer does not expect requests for late submission.
- **Lab Sheet 1** (clustering) – many students did it in the lab. These are **exam-style questions**: you get a data set and are asked to do certain things. There is no submission for the lab sheet, but doing the exercises gives hands-on experience.
- **Lab Sheet 2** – PCA and mixed-data dimension reduction (preparing data for PCA, doing PCA, **interpreting** PCA, MCA). Go through it next week.
- **Recordings:** there was an upload problem; Day 3 and Day 4 recordings are now uploaded, the others will follow.
- The **FAMD lecture note** (about 120 KB) is on the LMS and was shared in the Zoom chat.
- In the exam you won't need to memorise every command – you may be given the function name – but you must know how to use it. So practise.

2. Recap of Lab Sheet 1 (clustering)

The lab sheet used the fish data set (see Lecture 4 notes, Section 14, for the full code and results):

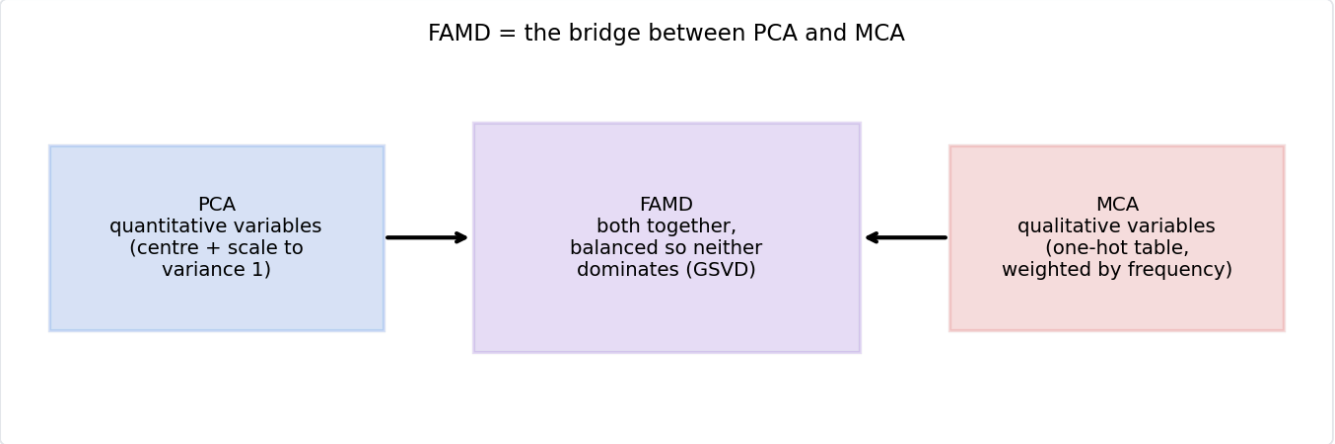
1. Remove the categorical variable (Species).
2. **Standardise** the data and look at the first few rows.
3. Compute the **Euclidean distance** and perform **hierarchical clustering**.
4. Look at the dendrogram – about **3 clusters**; draw cluster rectangles (`rect.hclust`).
5. `set.seed(123)` for reproducibility (any number works), then **k-means**; plot the result.
6. Compare the two methods with `table()`.

Exercises: add more species and traits; try another distance (Manhattan); use the **NbClust** package to find the optimal number of clusters automatically (no need to judge the dendrogram yourself); apply clustering to another data set such as iris. The lecturer showed the answers – NbClust gives the optimal number of clusters.

3. Why we need methods for mixed data

- Typical PCA works for **quantitative** data. If there are qualitative variables, we have to remove them and use only the quantitative ones.
- But what if we have **only** qualitative data, or a **significant portion** of the data is qualitative? Throwing it away loses information. We need a mechanism.
- There are special dimension-reduction methods beyond basic PCA:
  - **MCA** – Multiple Correspondence Analysis – for categorical data (Tutorial 1).
  - **FAMD** – Factor Analysis of Mixed Data – for **mixed** data (quantitative + qualitative). Today's topic.

4. FAMD – the idea and how it works



Factor Analysis of Mixed Data (FAMD) is a principal component method for analysing a data set that contains **both quantitative (continuous) and qualitative (categorical)** variables. Think of FAMD as the **"bridge"** between:

1. **PCA** – used for quantitative data;
2. **MCA** – used for qualitative data.

4.1 The core objective

Real-world data sets often mix data types, e.g. **Age, Income** (continuous) with **Gender, Occupation** (categorical). Standard PCA cannot handle categories, and MCA is not designed for continuous numbers. FAMD allows us to:

- **Visualise similarities** between individuals in a low-dimensional space;
- **Analyse the relationships** between all variables, regardless of type;
- **Reduce dimensionality** while keeping the maximum amount of "inertia" (variance).

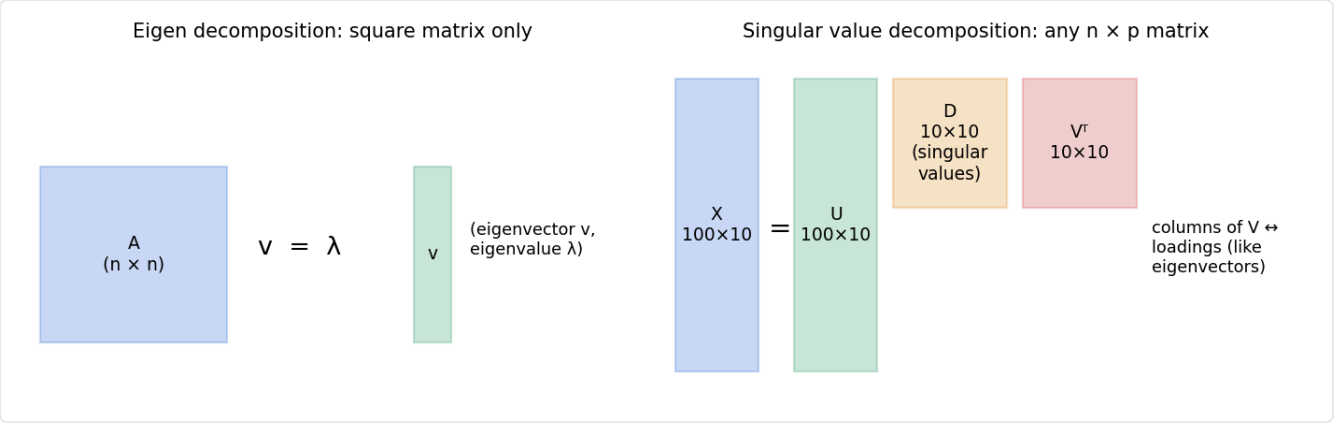
> **Job interview tip**
> In technical interviews for data scientist, data analyst or data engineer roles (and even internships), you may be asked about techniques like PCA and FAMD. You should know what FAMD is and when to use it.

4.2 The mechanism

FAMD **balances** the influence of both variable types so that one doesn't dominate the other.

- **Quantitative variables:** centred and scaled to unit variance (mean = 0, variance = 1) – i.e. standardised.
- **Qualitative variables:** turned into a **disjunctive data table** (one-hot encoding: one 0/1 column per category) and then scaled according to the **frequency** of each category.
- **Mathematical integration:** it uses the **Generalised Singular Value Decomposition (GSVD)**.

4.3 Eigen decomposition vs singular value decomposition



- **Eigen value decomposition:** for a **square** matrix A (n × n), we find a scalar λ (the **eigenvalue**) and a vector v (the **eigenvector**) such that **A v = λ v**. This works only for square matrices (rows = columns).
- In practice a data set is **not square**, e.g. 100 observations × 10 variables (a 100 × 10 matrix). So we use the **Singular Value Decomposition (SVD)**, which works for **any** n × p matrix: $X = U D V^T$. D holds the singular values; the columns of V give the directions (loadings).
- FAMD uses a **generalised** SVD, which adds weights for rows and columns (the category frequencies and variable scaling).

4.4 Inertia

> **Rule**
> The total inertia of a **quantitative** variable is **1**. The total inertia of a **qualitative** variable with **K levels** is **K − 1**.

Check with our examples (the eigenvalues add up to the total inertia):

Data	Variables	Total inertia	Dim 1 eigenvalue	Dim 1 %
Wine (5 columns)	3 quantitative (1 each) + Label (3 levels → 2) + Soil (4 levels → 3)	3 + 2 + 3 = 8	2.787	2.787 / 8 = 34.8%
Customers	Age 1 + Income 1 + Gender (3 → 2) + Region (2 → 1) + Subscription (2 → 1)	6	2.847	2.847 / 6 = 47.5%

Since this is an applied course, the lecturer did not go deep into the mathematics – the focus is on doing it in R.

5. FAMD in R – the wine example

The most robust package for FAMD is **FactoMineR**; **factoextra** is used for high-quality visualisations (we started using them last week).

5.1 Setup and data preparation

```
# install.packages(c("FactoMineR", "factoextra"))   # once
library(FactoMineR)
library(factoextra)
data(wine)
# select a mix of categorical and continuous columns
df <- wine[, c(1, 2, 3, 4, 5)]
str(df)          # check types: Factor and Numeric
```

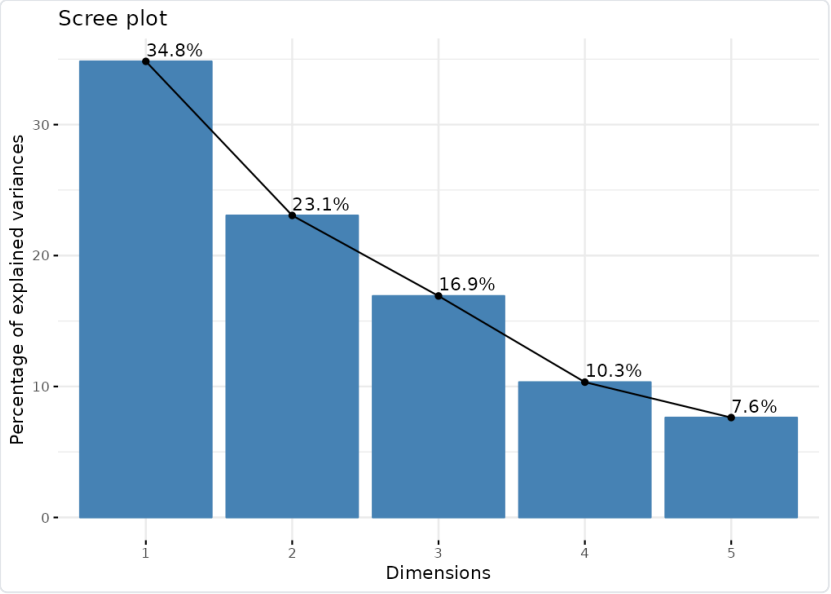
- `wine` (FactoMineR, `?wine`): **21 wines of the Val de Loire**. The first column is the **label of origin**, the second is the **soil**, the others are sensory descriptors.
- We keep 5 columns: **2 categorical** and **3 numeric**.

Column	Type	Details
Label	Factor (3)	Saumur 11, Bourgueuil 6, Chinon 4
Soil	Factor (4)	Reference 7, Env1 7, Env2 5, Env4 2
Odor.Intensity.before.shaking	num	e.g. 3.07, 2.96 …
Aroma.quality.before.shaking	num	e.g. 3.00, 2.82 …
Fruity.before.shaking	num	e.g. 2.71, 2.38 …

5.2 Running FAMD

```
res.famd <- FAMD(df, graph = FALSE)     # no graphs at this moment
res.famd                                # list of result objects (eig, var, ind, quali.var,
quanti.var)
# 1. Inspect eigenvalues (scree plot) — how much variance each dimension explains
fviz_eig(res.famd, addlabels = TRUE)
```

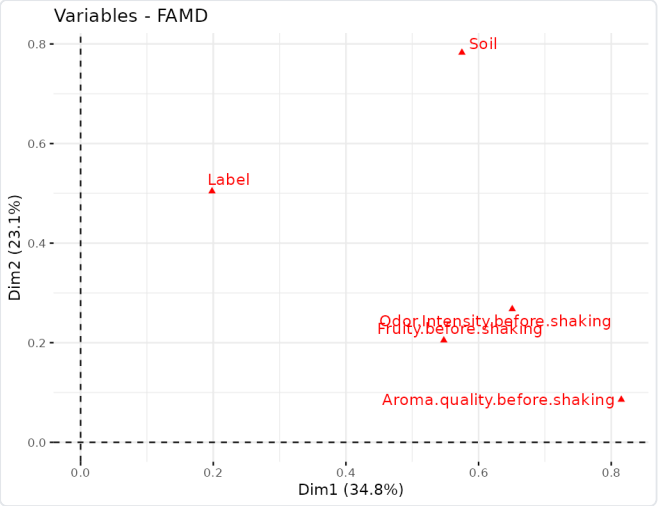
Dim	Eigenvalue	%	Cumulative %
1	2.787	34.8	34.8
2	1.845	23.1	57.9
3	1.353	16.9	74.8
4	0.827	10.3	85.1
5	0.610	7.6	92.8



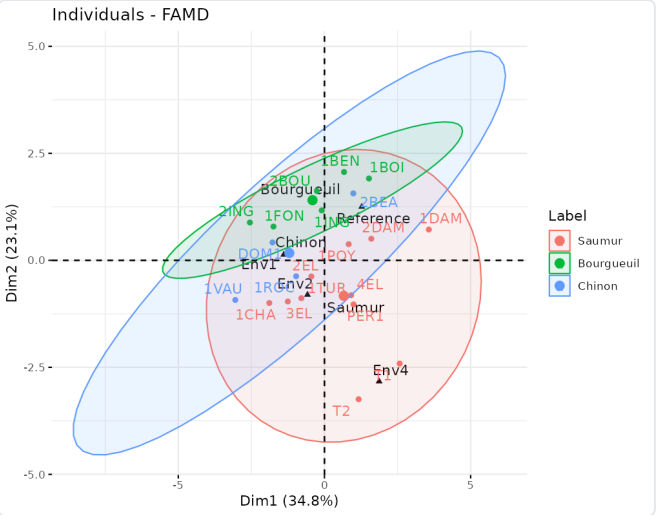
Dim 1 explains 34.8%, Dim 2 23.1% (the lecturer read "21.1%" in class; the output says 23.1%). Starting from 5 variables, you might keep **3** dimensions (an elbow), or 4, or 2 – there is no single clear answer; it is a little subjective.

5.3 Visualising the results

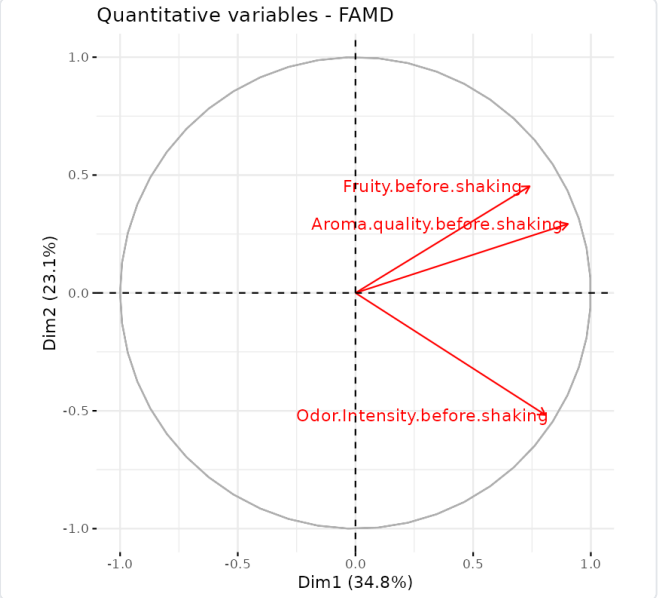
```
# 2. Graph of variables
fviz_famd_var(res.famd, repel = TRUE)
# 3. Graph of individuals, coloured by a categorical variable ("Label")
fviz_famd_ind(res.famd, habillage = "Label", addEllipses = TRUE, repel = TRUE)
```



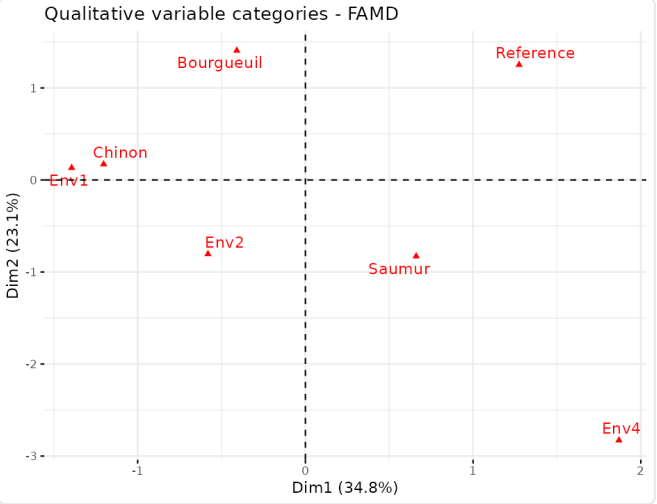
Variable plot (Dim 1 & 2)



Individuals coloured by Label, with ellipses



`choice = "quanti.var"`: correlation circle (arrows)



`choice = "quali.var"`: categories

- **Variable plot:** shows the "contribution" of each variable. In the default plot every variable is a point whose coordinates are its squared correlation (numeric) or η^2 (categorical) with the dimensions. Arrows (correlation circle) appear with `choice = "quanti.var"`.
- **Individual plot:** groups individuals by similarity. `habillage = "Label"` colours each wine by its label of origin; `addEllipses = TRUE` draws an ellipse around each group.
- **Reading it:** Dim 1 is built mostly by the three aroma/odour variables (all correlated positively with Dim 1: aroma quality 0.90, odour intensity 0.81, fruity 0.74) and Soil; Dim 2 mostly by **Soil** (42% contribution) and **Label** (27%). Soil "Env4" and "Reference" are far from the centre; Bourgueuil wines sit higher on Dim 2 than Saumur wines.

6. Interpretation guide and when to use FAMD

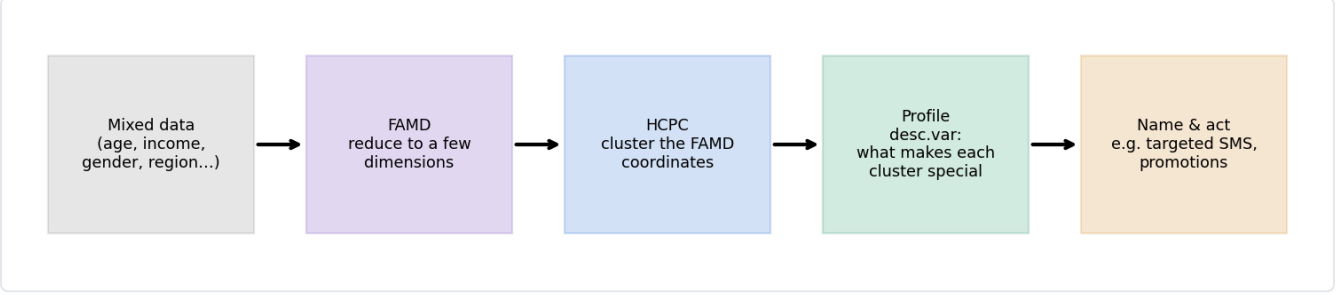
Output component	What it tells you
Eigenvalues	The amount of information kept by each dimension. Look for the "elbow" to decide how many new dimensions to keep.
Coordinates (coord)	Position of variables/individuals on the axes.
Contribution (ctr)	How much a variable contributes to a specific dimension. High ctr = high influence.
cos2 (quality)	How well a variable/individual is represented by the dimension. Values close to 1 are ideal.

> **Pro tip**
> In the `fviz_famd_var` plot, **quantitative variables that point in the same direction are positively correlated**. For **qualitative** variables, **categories that are close together** indicate that individuals often share those traits.

When to use FAMD

1. **Exploratory Data Analysis (EDA):** when there are too many variables to plot individually. As with PCA: 10 variables cannot be plotted, so we summarise them with 2 dimensions (e.g. Dim 1 = $c_1x_1 + c_2x_2 + \dots$) and plot those.
2. **Clustering pre-processing:** use FAMD dimensions as inputs for k-means or hierarchical clustering (**HCPC**) to handle mixed data and noise.
3. **Outlier detection:** individuals far from the centre of the factor map are often **multivariate outliers**.

7. Real-world story: customer segmentation



These are very applied problems. Companies (telecoms, insurance…) come to the Statistics department for consultancy, and many have their own analytics teams.

- **Telecom SMS promotion:** a provider has about **10 million** prepaid customers. Sending a promotional SMS (say Rs. 1 each) to everyone costs Rs. 10 million – not cost-effective, and the expected return may be less than that. Example: a 2 GB data offer for Rs. 400. Some people (e.g. many elderly people) have only a basic phone for calls, not a smartphone – sending them a data offer is a waste. So they **segment** customers using

features like **data usage** and current plans, and send the SMS only to the selected segments.

- **Pizza companies:** they send SMS/WhatsApp promotions, use **location-based** tracking through their app (a promotion when you pass their shop – "buy one get one free"), and periodic offers such as "no-cook Tuesday" – not sent to everyone, only to selected customer segments.
- **Churn:** a taco place called the lecturer: "we haven't seen you for 6 months – here is a promotion". To them he was a **churning customer**. **Churn** = a customer leaving / no longer using a brand or service (like an employee leaving a company). If you don't recharge a mobile SIM for 2 months, the company may contact you with offers. **Churn prediction** is a very famous problem; customer segmentation is used all the time.
- **In-house analytics teams:** Dialog, Mobitel, and Octave (the data analytics team of John Keells – Keells supermarkets, BYD cars, hotels…). Past students work there. In a big supermarket chain they can see almost in **real time** whether a promotion (e.g. on carrots) reached the expected revenue.
- Lesson: you can't just apply PCA and stop – real data contain categorical variables that you cannot ignore. So you need methods like FAMD.

8. Lab exercise: customer segmentation with FAMD + HCPC

This lab is self-contained: it generates a synthetic "customer research" data set with mixed types (Age, Income, Gender, Region, Subscription) and walks through the whole **FAMD → HCPC** pipeline.

8.1 Generate the synthetic data set

```
library(FactoMineR); library(factoextra); library(ggplot2)
set.seed(123)            # same seed → same data set for everyone
n <- 100
customer_data <- data.frame(
  Age    = c(rnorm(50, 25, 5), rnorm(50, 55, 8)),
  Income = c(rnorm(50, 40000, 5000), rnorm(50, 85000, 12000)),
  Gender = factor(sample(c("Male", "Female", "Non-binary"), n, replace = TRUE)),
  Region = factor(c(rep("Urban", 60), rep("Rural", 40))),
  Subscription = factor(c(
    sample(c("Basic", "Premium"), 50, replace = TRUE, prob = c(0.8, 0.2)),
    sample(c("Basic", "Premium"), 50, replace = TRUE, prob = c(0.2, 0.8)))))
summary(customer_data)
str(customer_data)
```

- The first 50 customers are **young** (age ~ N(25, 5)) with **lower income** (~40,000); the last 50 are **older** (~55) with **higher income** (~85,000).
- **Gender** is chosen completely at random (no pattern). **Region**: first 60 Urban, last 40 Rural. **Subscription**: first 50 mostly Basic (80%), last 50 mostly Premium (80%).
- `sample(..., replace = TRUE, prob = ...)` draws categories with the given probabilities.

Variable	Summary
Age	min 15.2, median 36.2, mean 40.7, max 72.5
Income	min 29,734, median 59,885, mean 62,098, max 123,892
Gender	Female 35, Male 33, Non-binary 32
Region	Rural 40, Urban 60
Subscription	Basic 49, Premium 51

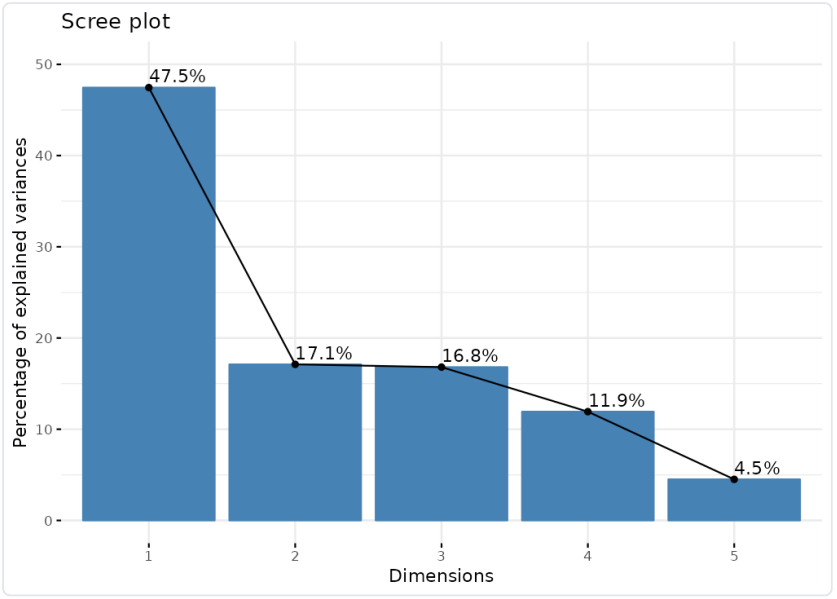
`str()`: 2 numeric variables and 3 factors. Before today we would have ignored the 3 categorical variables and done PCA on Age and Income only – losing the gender, region and subscription information.

8.2 Perform FAMD

```
res.famd <- FAMD(customer_data, ncp = 5, graph = FALSE)
fviz_eig(res.famd, addlabels = TRUE, ylim = c(0, 50))
```

- FAMD **automatically scales** the numeric variables and handles the factors – no need to one-hot encode or scale yourself.
- `ylim = c(0, 50)` sets the y-axis to 0–50% so the bars are "zoomed in" and differences are easier to see (0–100% would shrink them).

Dim	Eigenvalue	%	Cumulative %
1	2.847	47.5	47.5
2	1.027	17.1	64.6
3	1.009	16.8	81.4
4	0.716	11.9	93.3
5	0.270	4.5	97.8

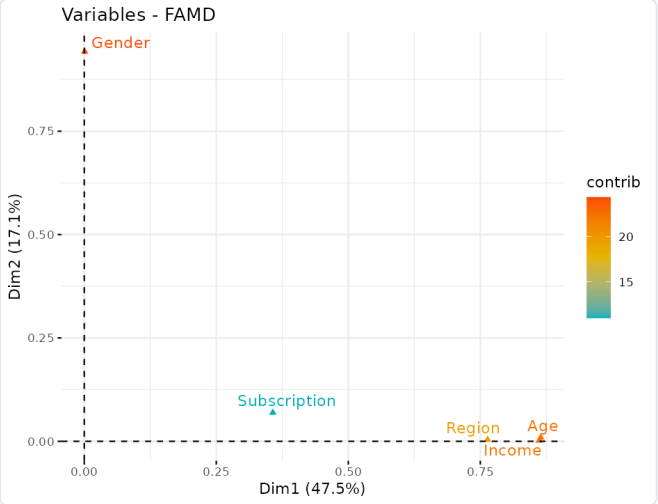


The elbow is right after Dim 1: one new dimension explains **47.5%** – we could go ahead with just one dimension.

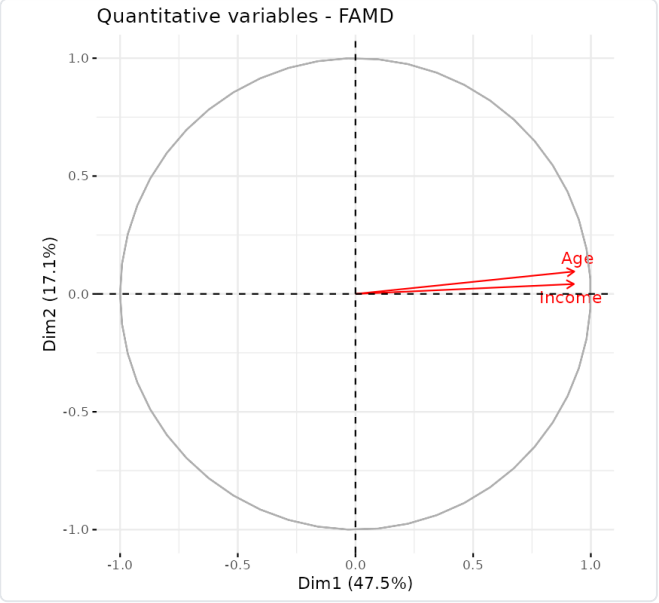
8.3 Analyse the variables

```
# Quantitative variables are represented by arrows, qualitative categories by points
fviz_famd_var(res.famd, repel = TRUE, col.var = "contrib",
              gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"))
```

- `col.var = "contrib"` colours variables by contribution; `gradient.cols` gives the colour scale (low → high). You don't need to memorise these arguments – know the function and check its help page.



Default variable plot – all variables are points

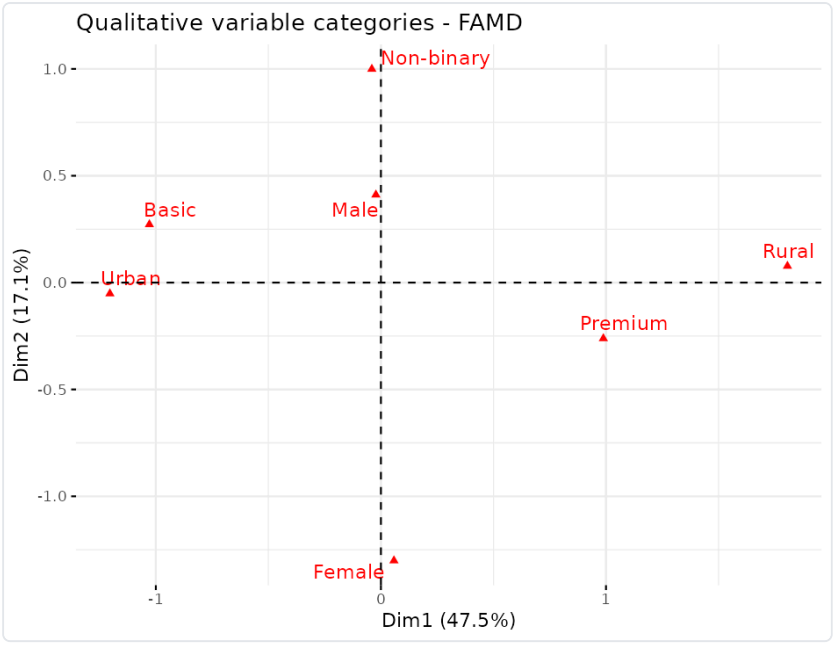


`choice = "quanti.var"` – arrows

> **The lecturer's doubt – answered**
> The comment in the lab code says quantitative variables are arrows and categories are points, and in class the lecturer said "I think it should be the other way around – we'll check". Result: in the **default** `fviz_famd_var()` plot (`choice = "var"`) **every variable (numeric or categorical) is a point** (triangle), placed by its squared correlation / η^2 with the dimensions. **Arrows** appear only in the correlation circle, `choice = "quanti.var"`. **Category points** appear with `choice = "quali.var"`.

Variable	Contrib Dim 1 %	Contrib Dim 2 %	Contrib Dim 3 %
Age	**30.4**	0.9	0.5
Income	**30.3**	0.2	0.1
Region	**26.8**	0.4	0.4
Subscription	12.5	6.8	0.2
Gender	0.0	**91.8**	**98.8**

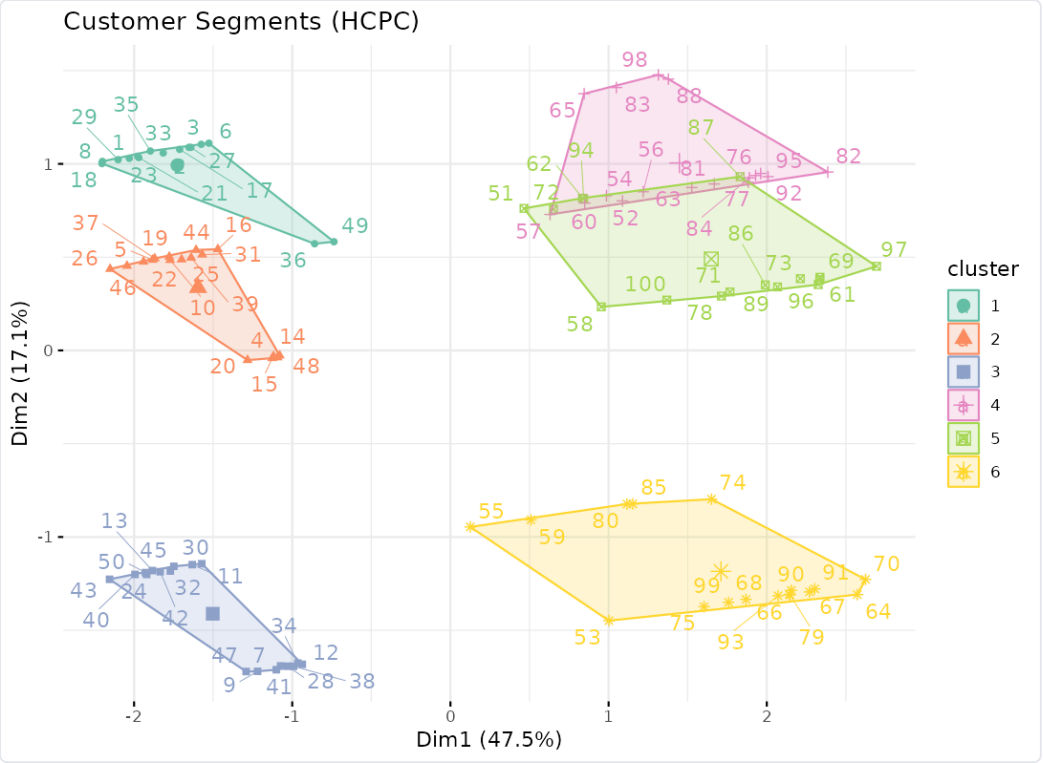
Correlations with Dim 1: Age 0.93, Income 0.93. Category coordinates on Dim 1: Rural +1.81, Premium +0.99 vs Urban −1.20, Basic −1.03. Gender builds Dims 2 and 3 on its own.

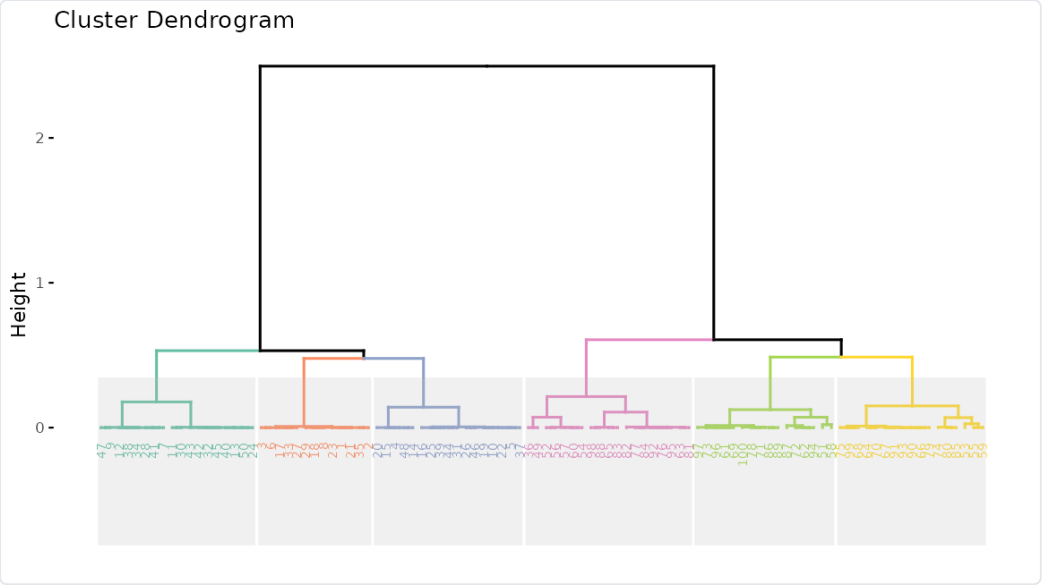


8.4 Clustering with HCPC

```
# Perform clustering on the FAMD coordinates
# 'nb.clust = -1' tells R to choose the optimal number of clusters automatically
res.hcpc <- HCPC(res.famd, nb.clust = -1, graph = FALSE)
# Visualise the clusters on the factor map
fviz_cluster(res.hcpc, repel = TRUE, show.clust.cent = TRUE, palette = "Set2",
             ggtheme = theme_minimal(), main = "Customer Segments (HCPC)")
```

- **HCPC** = Hierarchical Clustering on Principal Components. It runs (Ward) hierarchical clustering on the FAMD coordinates (not on the raw data), then improves the result with k-means.
- `nb.clust = -1` – like NbClust, R chooses the number of clusters automatically, so we don't have to decide from the dendrogram. (`0` lets you click on the tree; a positive number fixes it.)
- Result here: **6 clusters** with sizes 15, 17, 18, 17, 16, 17 (100 customers).





Extra: the HCPC tree, `fviz_dend(res.hcpc, rect = TRUE, rect_fill = TRUE)`

8.5 Interpreting the profiles – the "DNA" of each cluster

```
# 1. Which quantitative variables characterise each cluster?
#    Look for 'Mean in category' vs 'Overall mean'
res.hcpc$desc.var$quanti
# 2. Which categorical levels characterise each cluster?
#    Look for high percentages (e.g. 90% of Cluster 1 are "Urban")
res.hcpc$desc.var$category
```

How to read desc.var$quanti

- **Mean in category** vs **Overall mean**: is the cluster higher or lower than everyone?
- **v.test**: positive = significantly above average, negative = below; |v.test| > 2 is significant.

Cluster	Mean Age (overall 40.7)	Mean Income (overall 62,098)	v.test Age	v.test Income
1	25.2	39,057	−3.89	−3.87
2	25.0	39,791	−4.24	−4.04
3	25.3	37,456	−4.31	−4.62
4	57.5	80,141	+4.57	+3.27
5	54.8	87,434	+3.70	+4.42
6	56.1	88,938	+4.17	+4.86

In class the lecturer read this as "income is dominating" in each cluster. More precisely: **both Age and Income** are significant in **every** cluster – clusters 1–3 are **young, low-income** (negative v.tests) and clusters 4–6 are **older, high-income** (positive v.tests).

How to read desc.var$category

- **Cla/Mod**: % of people with this category who are in the cluster.
- **Mod/Cla**: % of the cluster that has this category ← the "high percentage" to look for.
- **Global**: % of everyone with this category. If Mod/Cla is much larger than Global, the category is typical of the cluster.

Cluster	Gender	Region	Subscription
1 (15)	100% Non-binary	100% Urban	87% Basic
2 (17)	100% Male	100% Urban	12 Basic / 5 Premium
3 (18)	100% Female	100% Urban	10 Basic / 8 Premium
4 (17)	100% Non-binary	71% Rural	76% Premium
5 (16)	100% Male	88% Rural	11 Premium / 5 Basic
6 (17)	100% Female	82% Rural	12 Premium / 5 Basic

E.g. cluster 1: Gender = Non-binary has Mod/Cla 100% vs Global 32% (v.test 5.98), Region = Urban 100% vs 60%. So "the higher the percentage, the more that category characterises the cluster".

9. Student discussion questions – answers

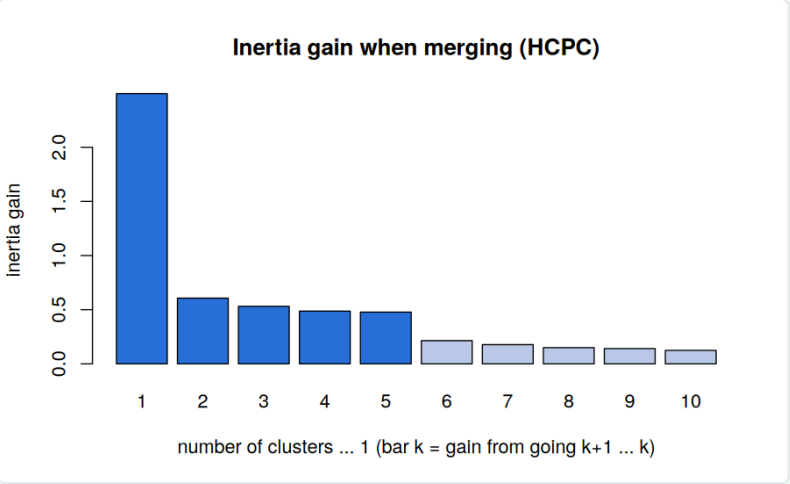
1. Dimension interpretation – which variables are most correlated with Dim 1? Wealth or age?
Age ($r = 0.93$), Income ($r = 0.93$), Region (η^2 0.76) and Subscription (η^2 0.36). Age and Income contribute almost equally (30.4% and 30.3%), so Dim 1 is **both** an age and a wealth dimension – a **"life stage / affluence"** dimension: older, richer, rural, premium customers (positive side) vs younger, lower-income, urban, basic customers (negative side). In this data age and income were generated together, so they cannot be separated.

2. Clustering logic – why that number of clusters? (Hint: res.hcpc$call$t$quot)

HCPC looks at the **loss (gain) of between-cluster inertia** when clusters are merged. `quot` = within-inertia with k clusters ÷ within-inertia with k − 1 clusters, for k = 3, 4, … 10:

k	3	4	5	6	7	8	9	10
quot	0.820	0.808	0.782	**0.727**	0.831	0.832	0.830	0.806

HCPC picks the k with the **smallest quot** – where adding one more cluster gives the biggest relative drop in within-cluster inertia, and after which further splitting helps much less. That is k = **6**. (By default HCPC searches between 3 and 10 clusters.)



3. Real-world application – names for the segments

Cluster	Suggested name
1	Young Urban Budget Explorers (non-binary)
2	Young Urban Male Starters (mostly Basic)
3	Young Urban Female Starters (Basic/Premium mix)
4	Established Rural Premium Users (non-binary)
5	Established Rural Male Premium Users
6	Established Rural Female Premium Users

In class this was a small competition: the boys named clusters 1–3 and the girls named clusters 4–6.

Think critically

Gender was generated **completely at random**, yet it splits the data into 3 groups on each side. This happens because Gender (3 levels, inertia 2) forms Dims 2 and 3 almost by itself (92% and 99% contribution), and HCPC uses all 5 dimensions. The real structure is **2 segments**: "young, low-income, urban, basic" vs "older, high-income, rural, premium". A business analyst should check whether splitting by gender is useful before acting – e.g. run HCPC on fewer dimensions (`FAMD(..., ncp = 1)` or `ncp = 2`) and compare.

10. Tutorial 2: FAMD on graduate employability

Learning outcomes: recognise when neither PCA nor MCA alone is enough; prepare mixed data; perform FAMD; interpret quantitative variables and categories on common dimensions; use contributions and cos2; treat an outcome as supplementary; extract FAMD coordinates as a reduced feature set. (Many of these results were already worked out in the Lecture 3 notes, Section 6; they are repeated here with the new parts.)

10.1 Data and variable types

```
dat <- read.csv("graduate_employability_mixed.csv", stringsAsFactors = TRUE)
str(dat); head(dat)
rownames(dat) <- dat$Graduate_ID
dat$Graduate_ID <- NULL
sapply(dat, class)
num_vars <- names(dat)[sapply(dat, is.numeric)];  num_vars
cat_vars <- names(dat)[sapply(dat, is.factor)];   cat_vars
```

Checkpoint 1 – answers

1. **8 quantitative**: GPA, Programming_Score, Statistics_Score, Communication_Score, Internship_Months, Portfolio_Projects, Weekly_SelfStudy_Hours, Hackathons.
2. **6 categorical**: Degree_Stream, Internship_Sector, English_Level, Teamwork_Level, Career_Preference, Employment_Status (5 active + 1 supplementary).
3. **PCA alone** is unsuitable because it can only use the 8 numeric variables; the 5 categorical predictors would be thrown away (or turned into fake numbers).
4. **MCA alone** is unsuitable because it needs categorical variables; the numeric scores would have to be cut into categories, losing information.
5. **One-hot + ordinary PCA**: each category becomes its own column, so a variable with many levels gets many columns and more total weight; the result also depends on how the 0/1 columns are scaled. So variables with many categories can have **disproportionate influence**.

10.2 Explore before reducing

```
summary(dat[num_vars])
lapply(dat[cat_vars], table)
colSums(is.na(dat))          # all 0 – no missing values
```

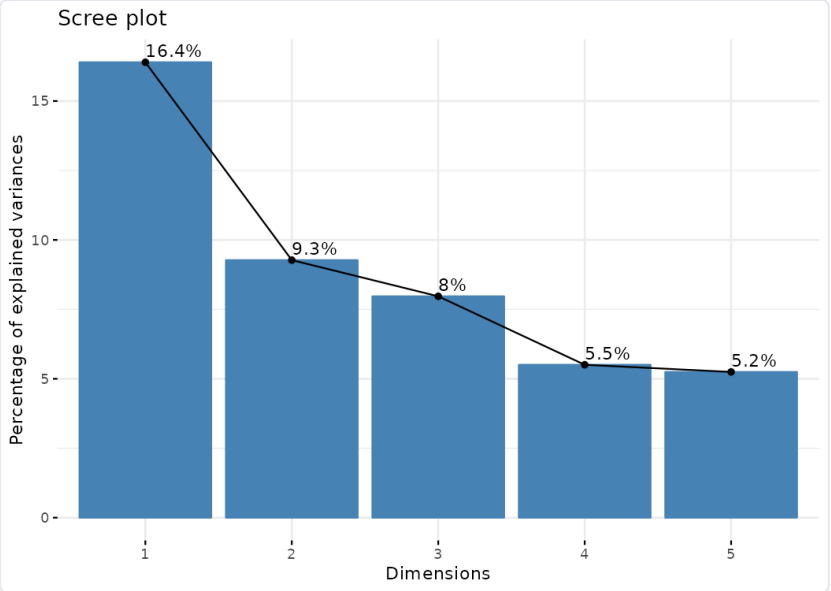
The data set was built without missing values so the focus is on FAMD. In a real analysis, the missing-data treatment must be justified before interpreting the factor map.

10.3 What FAMD does and running it

FAMD = PCA-like treatment of quantitative variables + MCA-like treatment of categorical variables in one balanced analysis.

```
sup_idx <- which(names(dat) == "Employment_Status")
res_famd <- FAMD(dat, sup.var = sup_idx, ncp = 5, graph = FALSE)
res_famd$eig
fviz_eig(res_famd, addlabels = TRUE)
```

`Employment_Status` is projected **after** the dimensions are built, so we can see whether the unsupervised structure lines up with employment, without using the outcome to build the axes.

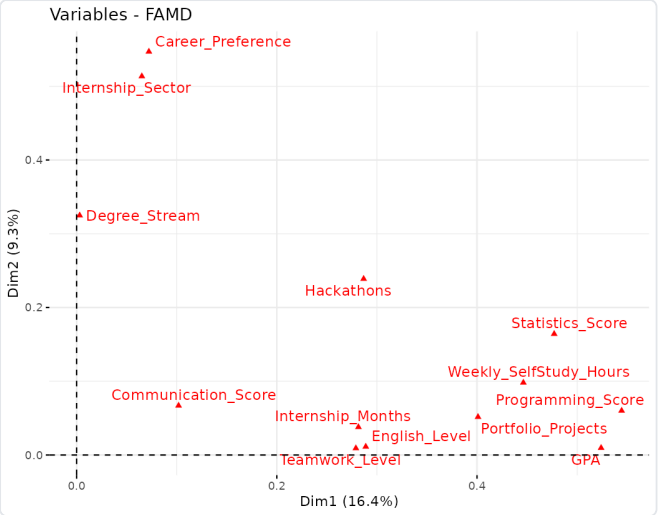


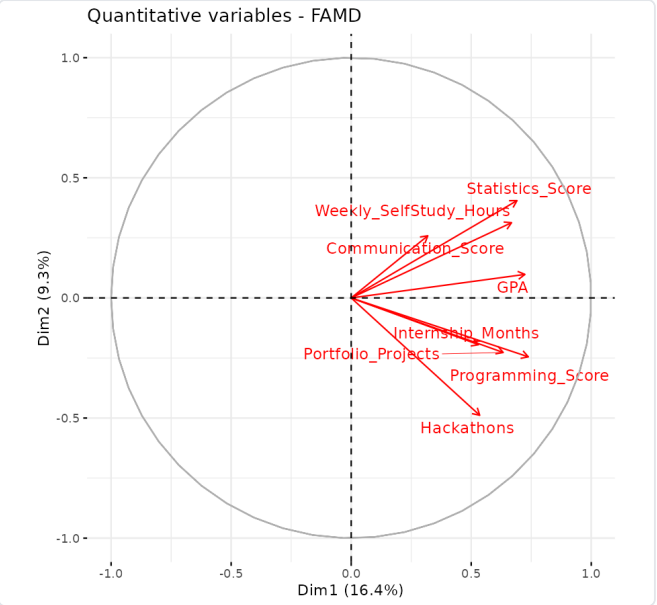
Checkpoint 2 – answers

1. Dims 1 and 2 summarise **25.7%** (16.4% + 9.3%).
2. A clear drop after Dim 1; after Dim 3 the values flatten (elbow around 3).
3. For **visualisation**: 2 dimensions.
4. For **clustering or prediction**: keep more (e.g. 3–5), because two dimensions hold only a quarter of the variability and useful information is in later dimensions.

10.4 Variable maps

```
fviz_famd_var(res_famd, repel = TRUE)                          # overall view
fviz_famd_var(res_famd, choice = "quanti.var", repel = TRUE)    # quantitative
res_famd$quanti.var$coord;  res_famd$quanti.var$contrib;  res_famd$quanti.var$cos2
```





How to interpret quantitative variables: pointing in a similar direction → positively associated; opposite sides → contrasting relationships with a dimension; near the origin → weakly represented on this plane; contribution → which variables create an axis; cos2 → how well a variable is represented.

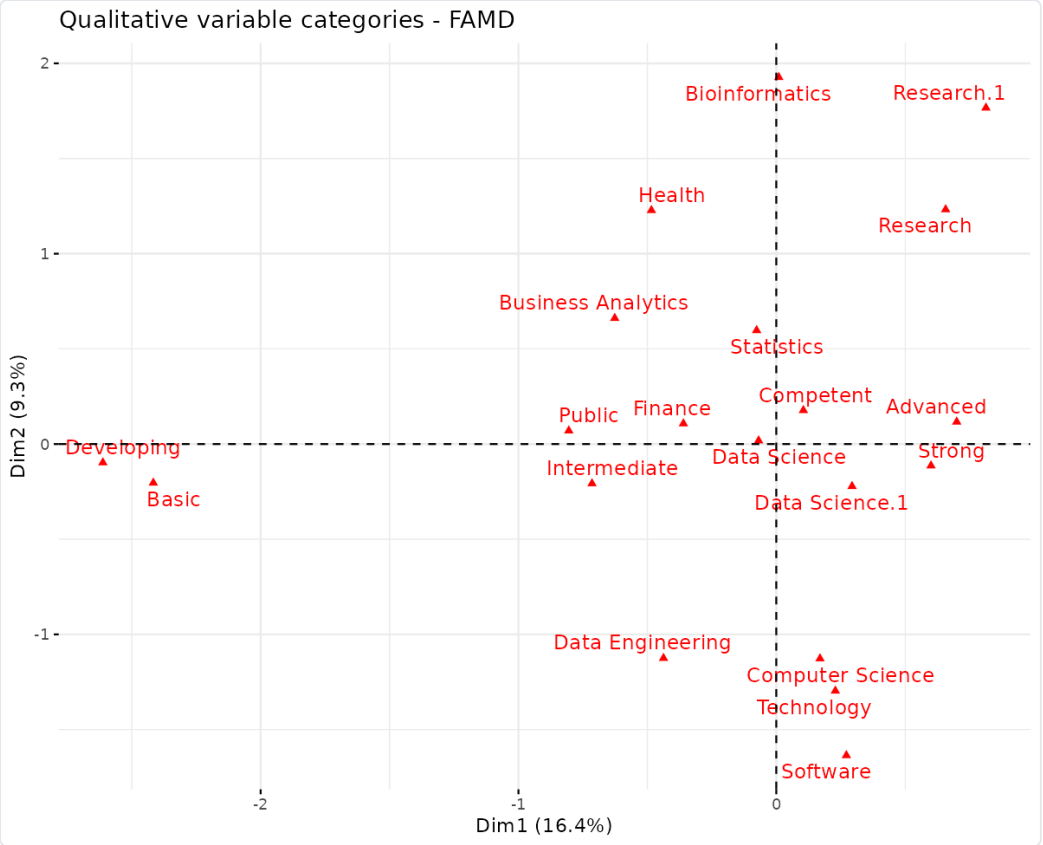
Variable	Coord Dim 1	Coord Dim 2	Contrib Dim 1 %	cos2 Dim 1 + 2
Programming_Score	**0.738**	−0.245	**14.4**	0.605
GPA	**0.724**	0.098	**13.9**	0.534
Statistics_Score	**0.691**	0.405	**12.7**	0.641
Weekly_SelfStudy_Hours	0.668	0.313	11.8	0.544
Portfolio_Projects	0.633	−0.227	10.6	0.453
Hackathons	0.535	−0.489	7.6	0.526
Internship_Months	0.531	−0.195	7.5	0.319
Communication_Score	0.319	0.259	2.7	0.169

Checkpoint 3 – answers (Dimension 1)

1. Largest absolute coordinates: Programming_Score (0.74), GPA (0.72), Statistics_Score (0.69).
2. Largest contributions: the same three (14.4%, 13.9%, 12.7%), then self-study hours and portfolio projects.
3. Programming_Score, Statistics_Score, Internship_Months and Portfolio_Projects are all **aligned** (all positive on Dim 1). On Dim 2, Statistics_Score goes up (+0.41) while the others go slightly down.
4. Name: **"Overall academic and technical preparation"** (or "technical preparation and portfolio intensity").

10.5 Categorical variables and categories

```
fviz_famd_var(res_famd, choice = "quali.var", repel = TRUE)
res_famd$quali.var$coord
res_famd$quali.var$contrib
```



Interpretation task – answers

- **Technically oriented profiles:** Internship_Sector = Technology, Career = Software and Data Engineering, Degree = Computer Science (negative side of Dim 2), together with more Hackathons and higher Programming scores.
- **Research-oriented profiles:** Career = Research ("Research.1"), Internship_Sector = Research and Health, Degree = Bioinformatics (positive side of Dim 2), with higher Statistics scores and self-study; Employment = Higher Studies is also here.
- **Business/communication-oriented profiles:** Career = Business Analytics, Internship_Sector = Finance / Public, higher Communication_Score – these are less clearly separated (they sit near the middle and load mainly on Dim 3), so don't force a strong story.
- **Weaker overall preparation:** English_Level = Basic, Teamwork_Level = Developing (far negative on Dim 1), where "Seeking" graduates also lie.

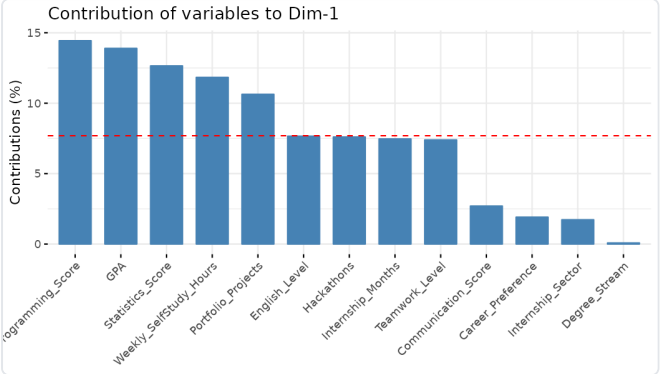
Some categories (e.g. Degree = Data Science, Intermediate English) are weakly represented on the first two dimensions.

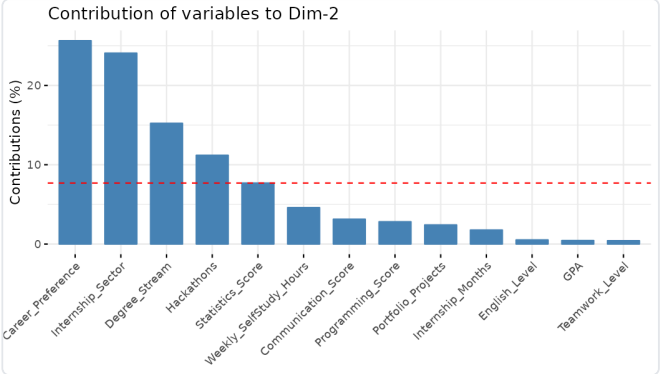
Duplicate category names

"Data Science.1" and "Research.1" are the Career_Preference categories (renamed because Degree_Stream and Internship_Sector already have those names).

10.6 Contributions

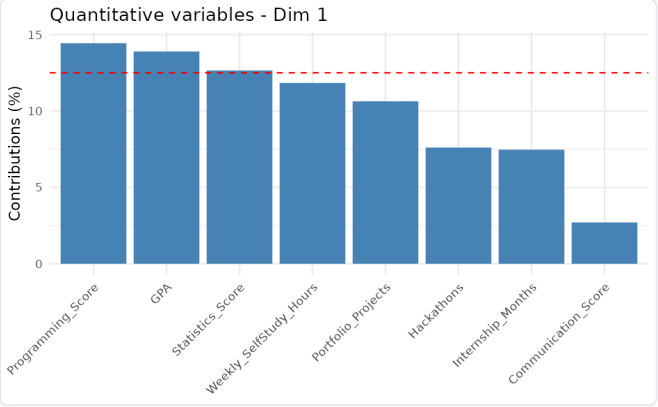
```
fviz_contrib(res_famd, choice = "var", axes = 1, top = 15)
fviz_contrib(res_famd, choice = "var", axes = 2, top = 15)
fviz_contrib(res_famd, choice = "quanti.var", axes = 1, top = 10)   # ERROR in this version
fviz_contrib(res_famd, choice = "quali.var",  axes = 1, top = 10)
```



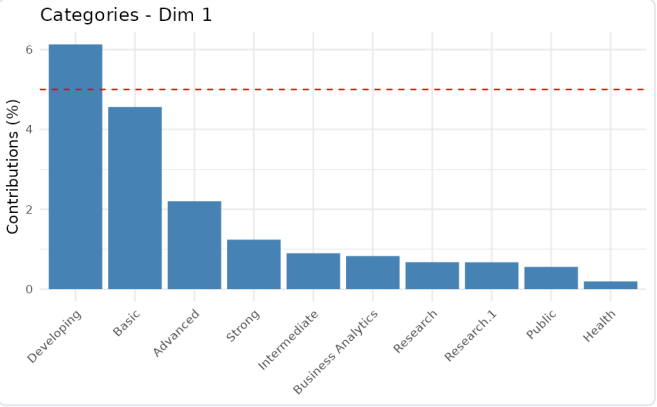


Code note

With factoextra 1.0.7, `fviz_contrib(res_famd, choice = "quanti.var", ...)` gives `Error in -nrow(X$group$Lg): invalid argument to unary operator`. Workaround: use `choice = "var"` (quantitative variables are included), or plot the numbers yourself: `barplot(sort(res_famd$quanti.var$contrib[,1], decreasing = TRUE), las = 2)`. The `"quali.var"` version works.



Quantitative contributions to Dim 1 (made from `$quanti.var$contrib`)



Category contributions to Dim 1

Top categories for Dim 1: Developing (6.1%), Basic (4.6%), Advanced (2.2%), Strong (1.2%).

10.7 Quality of representation (cos2)

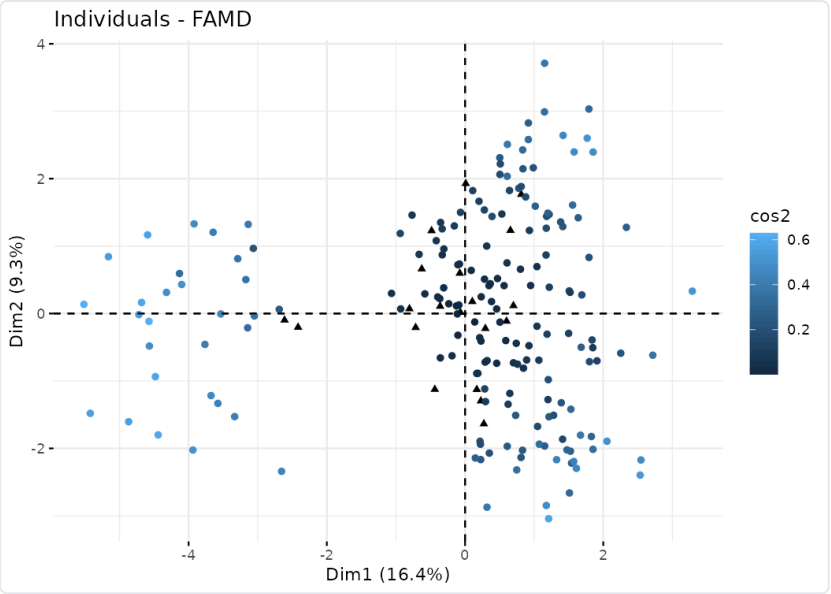
```
fviz_cos2(res_famd, choice = "var", axes = 1, top = 15)
quant_cos2_2d <- rowSums(res_famd$quanti.var$cos2[, 1:2]);  sort(quant_cos2_2d, decreasing
= TRUE)
qual_cos2_2d  <- rowSums(res_famd$quali.var$cos2[, 1:2]);   sort(qual_cos2_2d,  decreasing
= TRUE)[1:15]
```

Quantitative (cos2 Dim 1+2)	Categories (cos2 Dim 1+2)
Statistics_Score 0.641, Programming_Score 0.605, Weekly_SelfStudy_Hours 0.544, GPA 0.534, Hackathons 0.526, Portfolio_Projects 0.453, Internship_Months 0.319, Communication_Score 0.169	Technology 0.658, Developing 0.602, Advanced 0.555, Basic 0.488, Research.1 (career) 0.474, Research (sector) 0.425, Computer Science 0.405, Software 0.401, Strong 0.266, Business Analytics 0.258, Bioinformatics 0.212, Health 0.195, Data Engineering 0.181, Intermediate 0.154, Statistics 0.153

Communication_Score (0.17) is poorly represented on the first plane – it belongs mainly to Dim 3.

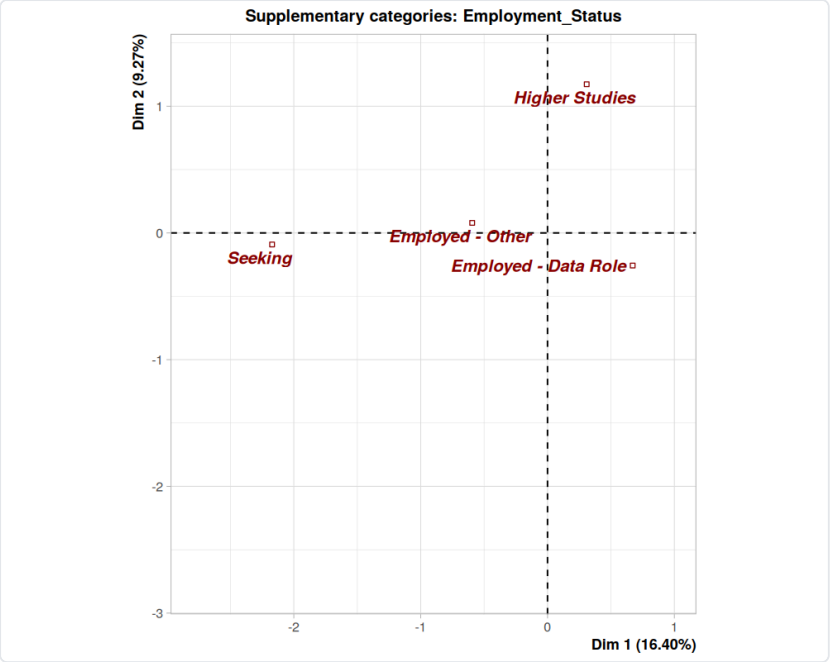
10.8 Individuals and reduced coordinates

```
fviz_famd_ind(res_famd, geom = "point", col.ind = "cos2")
graduate_coordinates <- as.data.frame(res_famd$ind$coord)
head(graduate_coordinates)
X_reduced <- graduate_coordinates[, 1:3]    # three FAMD dimensions instead of the full mixed data
```



10.9 Supplementary employment outcome

```
res_famd$quali.sup$coord
fviz_famd_var(res_famd, choice = "quali.sup", repel = TRUE)    # ERROR in current factoextra
plot(res_famd, choix = "ind", invisible = c("ind", "quali"))   # working alternative
```



Questions – answers

1. **Employed – Data Role:** the positive (strong-preparation) side of Dim 1 (coord 0.67, v.test 5.35), slightly towards the technical (negative) side of Dim 2.
2. **Seeking:** far on the negative side of Dim 1 (−2.17, v.test −6.42) – weaker preparation, Basic English, Developing teamwork.
3. **Higher Studies:** yes – high on Dim 2 (1.17, v.test 4.11), with Research career/sector, Bioinformatics and Health.
4. **Associations, not causes:** the data are observational; other factors (family, networks, economy, motivation) may affect both skills and employment, and the direction could also be reversed. The axes were built without the outcome, so we only see where outcomes lie.

10.10 Describe the dimensions

```
dimdesc(res_famd, axes = 1:2)
```

Labels from the actual output: **Dim 1 = "technical preparation and academic strength"** (all 8 scores positive; English Advanced, Teamwork Strong vs Basic, Developing); **Dim 2 = "research/statistical orientation versus technology/software orientation"** (Career & Sector Research, Bioinformatics, Statistics score vs Technology sector, Software, Computer Science, Hackathons).

10.11 Why not simply PCA after dummy-coding? (Checkpoint 4)

Thought experiment: English_Level has 3 categories, Career_Preference 5, Internship_Sector 5. One-hot encoding creates many binary columns (here 28 columns in total). An ordinary PCA of that expanded matrix depends on how the columns are scaled and can give more influence to variables with more levels. FAMD uses a weighting designed to balance them.

Demonstration (share of PC1 / Dim 1 coming from each variable):

Variable	Dummy-code + standardised PCA (PC1)	FAMD (Dim 1)
English_Level (3 dummy columns)	**16.2%** – the largest!	7.7%
Programming_Score	12.8%	14.4%
GPA	11.7%	13.9%
Teamwork_Level (3 dummy columns)	9.7%	7.4%

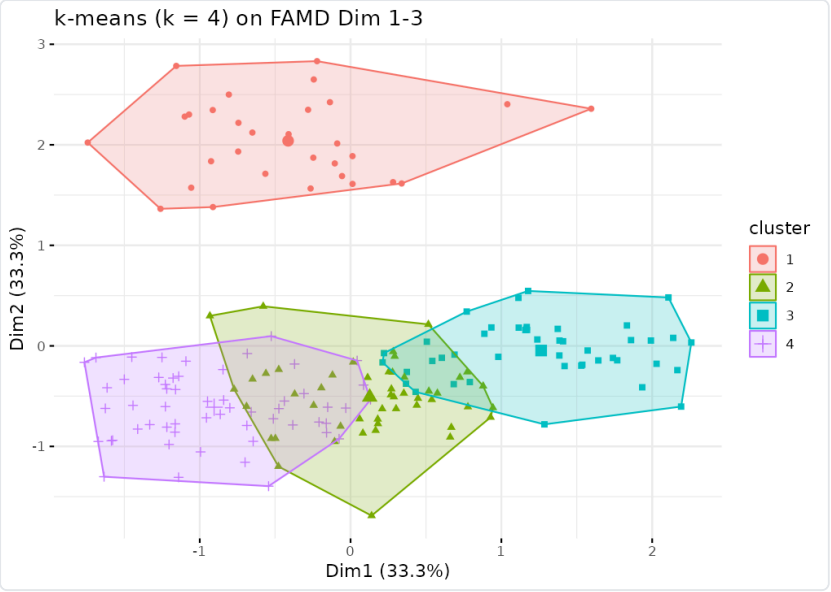
Checkpoint 4 – model answer (two sentences)
Dummy-coding every category and running ordinary PCA lets categorical variables gain influence just from their number of levels and from the arbitrary scaling of the 0/1 columns – here English_Level becomes the most important variable on PC1. FAMD standardises numeric variables and weights categories like MCA, so every variable contributes in a balanced way and the dimensions reflect real structure rather than coding choices.

10.12 Optional extension: FAMD features for clustering

```
coords <- as.data.frame(res_famd$ind$coord)
set.seed(123)
km <- kmeans(coords[, 1:3], centers = 4, nstart = 30)
table(km$cluster)                                  # 30 49 45 56
table(km$cluster, dat$Employment_Status)
```

Cluster (size)	Centre (Dim1, Dim2, Dim3)	Data Role	Other	Higher Studies	Seeking
1 (30)	−3.97, −0.19, 0.56	4	6	3	**17**
2 (49)	0.99, 1.61, 0.79	20	8	**15**	6
3 (45)	0.09, 0.27, −1.90	32	7	3	3
4 (56)	1.19, −1.52, 0.54	**47**	5	2	2

Cluster 1 = weaker preparation (most "Seeking"); cluster 2 = research-oriented (most "Higher Studies"); cluster 4 = strong technical (mostly "Data Role"); cluster 3 = middle / Dim 3 group.



Important

FAMD itself is not a clustering algorithm. Here it is used to create a reduced representation **before** clustering.

10.13 Mini-report – model answer (about 350 words)

Why FAMD? The data set combines eight quantitative skill measures (GPA, programming, statistics and communication scores, internship months, portfolio projects, self-study hours, hackathons) with five categorical descriptors (degree stream, internship sector, English level, teamwork level, career preference). PCA would ignore the categorical variables and MCA would need the scores to be cut into categories. FAMD analyses both types together and balances their influence.

How many dimensions? Dim 1 explains 16.4% and Dim 2 9.3% (25.7% together); the scree plot flattens after the third dimension (33.6%). I keep two dimensions for interpretation and plotting and three for clustering.

Dimension 1 measures overall technical and academic preparation: all quantitative variables correlate positively with it, led by Programming_Score (r = 0.74, 14.4%), GPA (0.72, 13.9%) and Statistics_Score (0.69, 12.7%). Advanced English and Strong teamwork lie on the positive side; Basic English and Developing teamwork on the negative side.

Dimension 2 contrasts research/statistical orientation (Research career and sector, Health, Bioinformatics, higher Statistics scores) with technology/software orientation (Technology sector, Software and Data Engineering careers, Computer Science, more hackathons). It is driven mainly by Career_Preference (25.6%), Internship_Sector (24.1%) and Degree_Stream (15.2%).

Most influential: quantitative – Programming_Score, GPA, Statistics_Score; categorical levels – Teamwork = Developing, English = Basic (Dim 1) and Technology sector, Research career, Software (Dim 2).

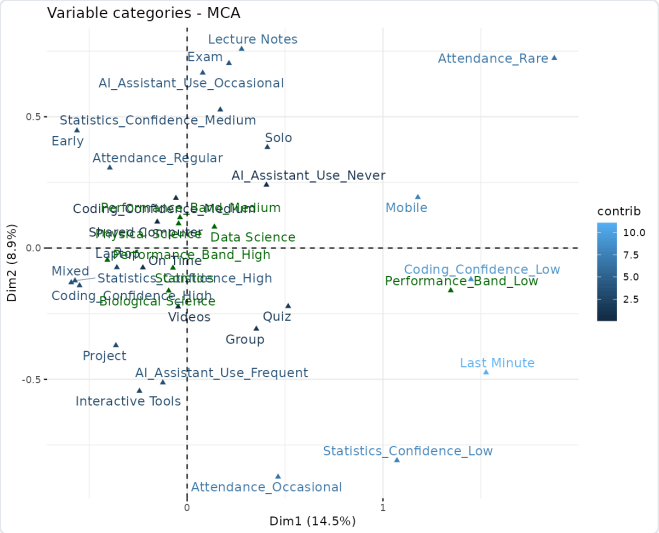
Supplementary outcome: Employed – Data Role lies on the strong side of Dim 1 (v.test 5.3), Seeking far on the weak side (v.test −6.4), and Higher Studies high on Dim 2 (v.test 4.1). These are associations, not causal effects.

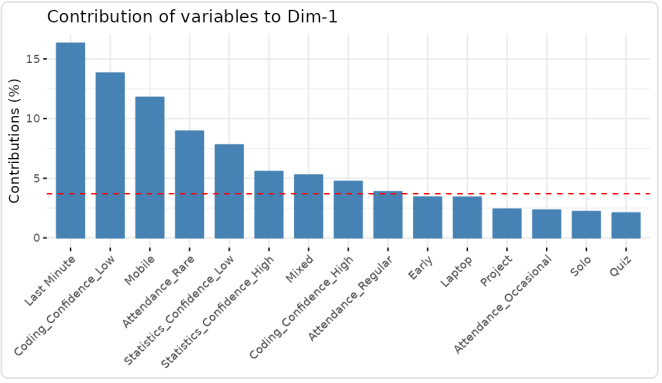
Limitations of two dimensions: (1) they retain only about a quarter of the variability, so some variables (e.g. Communication_Score, cos2 = 0.17) are poorly represented; (2) points close together on the plane may be far apart in later dimensions, so distances and groupings on the map can mislead.

Include: the scree plot, the quantitative-variable map, the qualitative-category map and one contribution plot.

11. Tutorial 1 (MCA) – short recap

Tutorial 1 (MCA on `student_learning_profiles_categorical.csv`) was fully worked out in the **Lecture 3 notes, Sections 3–4** (all checkpoints, outputs, plots and the mini-report). Key results:





Item	Result
Active / supplementary	9 learning-behaviour variables / Programme and Performance_Band
Eigenvalues (first 5)	0.291, 0.178, 0.156, 0.148, 0.136 (14.5%, 8.9%, 7.8%, 7.4%, 6.8%); 18 dimensions in total
Dim 1 top categories	Last Minute, Coding_Confidence_Low, Mobile, Attendance_Rare, Statistics_Confidence_Low → "engaged vs at-risk learning behaviour"
Dim 2 top categories	Attendance_Occasional, Lecture Notes, AI_Occasional, Exam, AI_Frequent → "traditional vs interactive learning style"
Best represented (cos2 plane)	Last Minute 0.57, Attendance_Regular 0.46, Coding_Confidence_Low 0.44
Supplementary	Performance Low on the at-risk side (v.test 7.6), High on the engaged side (−5.0); Programme not separated
Indicator vs Burt	Burt eigenvalues = squares of indicator eigenvalues; same directions and interpretation

12. The lecture R file (recap) and summary

The R file shared with this lecture contains the code already explained in the Lecture 3 notes: reading the MCA data (and `read.csv("data2.txt", sep = ";")` for semicolon-separated files), `?mca` vs `?MCA` (R is case-sensitive), the full MCA workflow with `graph = FALSE` and `graph = TRUE`, the Part B Task 7 dimension names ("Dimension 1 – Learning Behaviour and Study Habits", "Dimension 2 – Learning Preferences and Engagement"), the FAMD workflow for Part C (Tasks 7, 8: extract the first three coordinates for clustering), and the start of cluster analysis (five genes, single and complete linkage). Remember the two code fixes: `fviz_famd_var(..., choice = "quali.sup")` → use `plot(res_famd, choix = "ind", invisible = c("ind","quali"))`, and `fviz_contrib(..., choice = "quanti.var")` for FAMD → use `choice = "var"` or a barplot.

Key points

- PCA → quantitative; MCA → categorical; **FAMD → mixed** (the bridge).
- FAMD standardises numeric variables, one-hot codes and frequency-weights categories, and uses **GSVD**. Inertia: numeric variable = 1, categorical with K levels = K − 1.
- Eigen decomposition ($A\ v = \lambda\ v$) needs a square matrix; SVD works for any data matrix.
- Read eigenvalues (elbow), coordinates, contributions (influence) and cos2 (quality).
- Uses: EDA, pre-processing for clustering (FAMD → **HCPC**), outlier detection.
- HCPC: `nb.clust = -1` picks k automatically using the inertia ratio `quot`; profile clusters with `desc.var$quanti` (mean in category vs overall mean) and `desc.var$category` (Mod/Cla vs Global).
- Customer segmentation and churn are everyday tasks in telecom, retail and food companies.
- Always think critically: a random variable (like Gender here) can create clusters that are not meaningful.

Test yourself

1. Why is FAMD called a bridge between PCA and MCA?
2. How does FAMD treat quantitative and qualitative variables?
3. A data set has 4 numeric variables and 2 factors with 3 and 5 levels. What is the total inertia? (Answer: 4 + 2 + 4 = 10)
4. Why can't we use eigen decomposition directly on a 100 × 10 data matrix?
5. Name three uses of FAMD.
6. What does `nb.clust = -1` do in `HCPC()`?
7. In `desc.var$category`, what do Mod/Cla and Global mean?
8. Why might the 6 customer clusters be misleading?
9. What is customer churn? Give an example.
10. Why is "dummy-code everything + PCA" not a good substitute for FAMD?